

Yuexin ZHANG

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Employment

Center for Astrophysics, Harvard & Smithsonian, US NWO Rubicon Fellow	Nov 2023–Oct 2025
Kapteyn Astro. Institute, University of Groningen, NL NWO Rubicon Fellow	Oct 2023–Oct 2025

Education

Kapteyn Astro. Institute, University of Groningen, NL PhD in Astronomy. Advisor: Mariano Méndez and Diego Altamirano	Oct 2019–Sep 2023
Department of Physics, Fudan University, CN B.S. (<i>Honor</i>) in Physics. Advisor: Cosimo Bambi	Sep 2015–June 2019
Hamburg University, DE Summer Exchange Student	Jul 2018

Previous Academic Positions

University of Southampton, UK Visiting Researcher. PI: Diego Altamirano	Feb 2023
Institute of High-Energy Physics, CAS, CN Visiting Researcher. PI: Shuang-Nan Zhang and Jin-Lu Qu	Mar 2021–Mar 2022
Shanghai Astronomical Observatory, CAS, CN Summer Visiting Student. PI: Wenfei Yu	Jun 2019–Sep 2019

Member

<i>IXPE</i> Science Team	2024–present
<i>NICER</i> Science Team	2023–present
<i>Insight</i> -HXMT Science Team	2021–2022

Professional Service

Reviewer for ApJ	2025
Panelist for NASA proposals	2024–2025

Honors and Awards

NWO Rubicon Fellowship (153k EUR)	2023–2025
<i>“New light on the origin of X-ray variability in Galactic accreting black holes”</i>	
CSC and UoG Scholarship (96k EUR)	2019–2023
<i>“Spectral variability of compact X-ray sources”</i>	
Shanghai Outstanding Graduate	2019
Wangdao Scholar (named after the former president of Fudan)	2019
Fudan’s Undergraduate Research Opportunities Program (10k CNY)	2018–2019
<i>“Testing general relativity using X-ray reflection spectroscopy”</i>	

Observations

PI:

- [1] *Chandra* Cycle 27, “A multi-wavelength, multi-dimensional campaign of the disk wind in *GRS 1915+105*”, 50 ks
- [2] *Magellan* 2025B, “The Mass and Spin of Two Black Hole X-ray Binaries”, 3 half-nights
- [3] *NuSTAR* Cycle 11, “Can power color hue efficiently capture the next transition in *GX 339-4* with *NuSTAR-NICER*?”, 60 ks (74k USD)
- [4] *NuSTAR* Cycle 11, “A multi-wavelength, multi-dimensional view of *GRS 1915+105* in its wind phase”, 40 ks (60k USD)
- [5] *NICER* Cycle 7, “A multi-wavelength, multi-dimensional view of the accretion wind in *GRS 1915+105*”, 50 ks (39k USD)
- [6] *NICER* DDT on V4641 Sgr, 30 ks
- [7] *Magellan* 2024B, “The mass and spin of *Swift J1727.8–1613*”, 4 half-nights
- [8] *NICER* DDT on *GX 339–4*, 10 ks
- [9] *Insight-HXMT* AO–6, “Unravelling the origin of fast time variability in X-ray binaries with *Insight-HXMT*”, 3.6 ms
- [10] *Insight-HXMT* AO–5, “On the QPO properties of a black-hole X-ray binary”, 10 ks

Co-I:

- [1] *NuSTAR* Cycle 11, “Observation of a black hole X-ray binary in the dim hard state”, PI: König, 100 ks
- [2] *NICER* Cycle 7, “A legacy of turtles in *NICER* Cycle 7: Amassing a comprehensive spectral-timing archive of outbursting black holes”, PI: Steiner, 120 ks
- [3] *NICER* Cycle 7, “A long look at the hard state of *LMC X-3* with *NICER*”, PI: Steiner,

30 ks

- [4] *NICER* Cycle 7, “Monitoring a transient black hole X-ray binary in the soft-to-hard transition”, PI: König, 60 ks
- [5] *IXPE* Cycle 2, “A test of disk-jet misalignment and black hole spin measurement for the X-ray binary *H1743–322*”, PI: Steiner, 320 ks
- [6] *IXPE* Cycle 2, “First polarimetric look into the heartbeat state of accreting black holes with *4U 1630–47*”, PI: Steiner, 300 ks
- [7] *NuSTAR* Cycle 10, “Slim-disk spectro-polarimetry with *LMC X-3*”, PI: Steiner, 120 ks
- [8] *IXPE* Cycle 1, “Evolving slim-disk polarization with *LMC X-3*”, PI: Steiner, 1.2 ms

Conferences and Talks

- [1] Conference, 21st Meeting of the High Energy Astrophysics Division, Horseshoe Bay, US
7–12 Apr 2024
“The evolution of the high-frequency variability in the black hole candidate *GRS 1915+105* as seen by *RXTE* (Poster)”
- [2] Seminar, Massachusetts Institute of Technology, USA 21 Feb 2024
“A *NICER* view of the corona through Comptonization modeling of the QPO in bright black-hole X-ray binaries”
- [3] Seminar, Wuhan University, CN 8 Oct 2023
“The accretion-ejection coupling in Galactic accreting black holes”
- [4] Seminar, University of Amsterdam, NL 13 Sep 2023
“The accretion-ejection coupling in Galactic accreting black holes”
- [5] Seminar, Yunnan Observatory, CN 14 Aug 2023
“The accretion-ejection coupling in Galactic accreting black holes”
- [6] Seminar, Fudan University, CN 1 Aug 2023
“The accretion-ejection coupling in Galactic accreting black holes”
- [7] Seminar, Institute of High-Energy Physics, CN 27 Jul 2023
“The accretion-ejection coupling in Galactic accreting black holes”
- [8] Conference, The First Vasto Accretion Meeting, Vasto, IT 19–23 Jun 2023
“The corona-jet evolution of black-hole X-ray transients *MAXI J1535–571* and *MAXI J1820+070* (Poster)”
- [9] Conference, 10th Microquasar Workshop, Crete, GR 22–26 May 2023
“The jet-like corona of black-hole X-ray transient from the *HIMS* to the *SIMS*”
- [10] Conference, Nederlandse Astronomen Conferentie 2023, Leeuwarden, NL 15–17 May 2023
“Corona as the energy reservoir for radio jet: a case study of *GRS 1915+105* (Poster)”

- [11] Conference, NOVA Network NW3, Amsterdam, NL 19 Jan 2023
“The jet-like corona of black-hole X-ray transients from the HIMS to the SIMS”
- [12] Conference, 44th COSPAR Scientific Assembly, Athens, GR 16–24 Jul 2022
“Corona evolution of MAXI J1535–571 revealed by type-C quasi-periodic oscillations observed with Insight-HXMT”
“The evolution of the high-frequency variability in GRS 1915+105 as seen by RXTE (Poster)”
- [13] Conference, Black Hole Accretion under the X-ray Microscope, Madrid, ES 14–17 Jun 2022
“The evolution of the corona in MAXI J1535–571 through type-C quasi-periodic oscillations with Insight-HXMT (Poster)”
- [14] Conference, China Astronomy Annual Meeting (online), Nanchong, CN 2–6 Dec 2021
“Mapping the Comptonization region of black holes up to 100 keV through quasi-periodic oscillations in the intermediate state with Insight-HXMT”

References

Prof. Mariano Méndez: mariano@astro.rug.nl
 Dr. James (“Jack”) Steiner: james.steiner@cfa.harvard.edu
 Prof. Diego Altamirano: d.altamirano@soton.ac.uk
 Prof. Shuang-Nan Zhang: zhangsn@ihep.ac.cn
 Prof. Cosimo Bambi: bambi@fudan.edu.cn

Publications

First and second author:

- [1] **Zhang, Y.**, Méndez, M., Motta, S. et al. (2024). *A systematic study of the high-frequency bump in the black-hole low-mass X-ray binary GX 339–4*. Monthly Notices of the Royal Astronomical Society, 527(3), 5638-5648.
- [2] **Zhang, Y.**, Méndez, M., García, F. et al. (2023). *A NICER look at the jet-like corona of MAXI J1535–571 through type-B quasi-periodic oscillation*. Monthly Notices of the Royal Astronomical Society, 520(4), 5144-5156.
- [3] **Zhang, Y.**, Méndez, M., García, F. et al. (2022). *The evolution of the high-frequency variability in the black hole candidate GRS 1915+105 as seen by RXTE*. Monthly Notices of the Royal Astronomical Society, 514(2), 2891-2901.
- [4] **Zhang, Y.**, Méndez, M., García, F. et al. (2022). *The evolution of the corona in MAXI J1535-571 through type-C quasi-periodic oscillations with Insight-HXMT*. Monthly Notices of the Royal Astronomical Society, 512(2), 2686-2696.
- [5] Tripathi, A., **Zhang, Y.**, Abdikamalov, A. B. et al. (2021). *Testing general relativity*

with NuSTAR data of galactic black holes. The Astrophysical Journal, 913(2), 79.

- [6] **Zhang, Y.**, Abdikamalov, A. B., Ayzenberg, D. et al. (2019). *Tests of the Kerr hypothesis with GRS 1915+105 using different RELXILL flavors.* The Astrophysical Journal, 884(2), 147.
- [7] **Zhang, Y.**, Abdikamalov, A., Ayzenberg, D. et al. (2019). *About the Kerr nature of the stellar-mass black hole in GRS 1915+105.* The Astrophysical Journal, 875 (1), 41.
- [8] **Zhang, Y.**, Zhou, M., & Bambi, C. (2018). *Iron line spectroscopy of black holes in asymptotically safe gravity.* The European Physical Journal C, 78 (5), 376.

Co-author:

- [1] Fan, N., Steiner, J., Bambi, C. et al., **incl. Zhang, Y.** (2024). *NICER spectral and timing analysis of 4U 1630–47 and its heartbeat state.* The Astrophysical Journal, 984(1), 31.
- [2] Alabarta, K., Méndez, M., García, F. et al., **incl. Zhang, Y.** (2024). *Geometry of the Comptonization region of MAXI J1348–630 through type-C quasi-periodic oscillations with NICER.* The Astrophysical Journal, 980(2), 251.
- [3] Zhang, Z., Bambi, C., Liu, H. et al., **incl. Zhang, Y.** (2024). *A variable ionized disk wind in MAXI J1803–298 revealed by NICER.* The Astrophysical Journal, 975(1), 22.
- [4] Podgorny, J., Svoboda, J., Dovciak, M. et al., **incl. Zhang, Y.** (2024). *Recovery of the X-ray polarisation of Swift J1727.8–1613 after the soft-to-hard spectral transition.* Astronomy & Astrophysics, 686, L12.
- [5] Jin, P., Zhang, G., **Zhang, Y.** et al. (2024). *The bright black hole X-ray binary 4U 1543–47 during 2021 outburst - I. A clear state transition from super-Eddington to sub-Eddington accretion revealed by Insight-HXMT.* Monthly Notices of the Royal Astronomical Society, 530(1), 929-946.
- [6] Svoboda, J., Dovciak, M., Steiner, J. et al., **incl. Zhang, Y.** (2024). *Dramatic drop in the X-ray polarization of Swift J1727.8–1613 in the soft spectral state.* The Astrophysical Journal Letters, 966(2), L35.
- [7] Zhao, S., Tao, L., Li, P. et al., **incl. Zhang, Y.** (2024). *The bright black hole X-ray binary 4U 1543–47 during 2021 outburst: a thick accretion disk inflated by high luminosity.* Astronomy & Astrophysics, 685, A42.
- [8] Zhang, L., Méndez, M., García, F. et al., **incl. Zhang, Y.** (2023). *Type-A quasi-periodic oscillation in the black hole transient MAXI J1348–630.* Monthly Notices of the Royal Astronomical Society, 526(3), 3944-3950.
- [9] Ma, R., Méndez, M., García, F. et al., **incl. Zhang, Y.** (2023). *Variable corona during the transition from type-C to type-B quasi-periodic oscillations in the black hole X-ray binary MAXI J1820+070.* Monthly Notices of the Royal Astronomical Society, 525(1),

854-875.

- [10] Yang, Z. X., Zhang, L., Zhang S. N. et al., **incl. Zhang, Y.** (2023). *Fast transitions of X-ray variability in the black hole transient GX 339–4: comparison with MAXI J1820+070 and MAXI J1348–630*. Monthly Notices of the Royal Astronomical Society, 521(3), 3570-3584.
- [11] García, F., Karpouzas, K., Méndez, M. et al., **incl. Zhang, Y.** (2022). *The evolving properties of the corona of GRS 1915+105: a spectral-timing perspective through variable-Comptonization modelling*. Monthly Notices of the Royal Astronomical Society, 513(3), 4196-4207.
- [12] Liu, H., Fu, Y., Bambi, C. et al., **incl. Zhang, Y.** (2022). *The disk wind in GRS 1915+105 as seen by Insight-HXMT*. The Astrophysical Journal, 933(2), 122.
- [13] Yang, Z. X., Liang, Z., Bu, Q. C. et al., **incl. Zhang, Y.** (2022). *The accretion flow geometry of MAXI J1820+070 through broadband noise research with Insight-HXMT*. The Astrophysical Journal, 932(1), 7.
- [14] Méndez, M., Karpouzas, K., García, F. et al., **incl. Zhang, Y.** (2022). *Coupling between the accreting corona and the relativistic jet in the microquasar GRS 1915+105*. Nature Astronomy, 6(5), 577-583.
- [15] Karpouzas, K., Méndez, M., García, F. et al., **incl. Zhang, Y.** (2021). *A variable corona for GRS 1915+105*. Monthly Notices of the Royal Astronomical Society, 503(4), 5522-5533.
- [16] Abdikamalov, A. B., Ayzenberg, D., Bambi, C. et al. **incl. Zhang, Y.** (2021). *Implementation of a radial disk ionization profile in the relxill_nk model*. Physical Review D, 103(10), 103023.
- [17] Liu, H., Ji, L., Bambi, C. et al., **incl. Zhang, Y.** (2021). *Testing evolution of LFQ-POs with mass accretion rate in GRS 1915+105 with Insight-HXMT*. The Astrophysical Journal, 909(1), 63.